IDS - UNIT-2

DATA QUICK Analysis.

↓ Quick Analysis tool is a feature that helps you quickly apply common data analysis tasks like formatting, charts, totals, tables and sparklines to your selected data with just a few clicks.
→ It acts like a shortcut menu.

→ Benifits:
① Save time ② Easy Data Visualization.
③ Simplified data analysis.

⇒ Components of quick Analysis:
① Formatting: Apply conditional formating like data bars, color scales & icon sets.
→ Visualize data trends with color-coded functions.

② Charts: Provides option to insert charts like column, line, Pie, & Bar charts.
→ Quickly visualize data patterns through charting.

③ Totals: Perform quick calculations like sum, average, count, % total & running tool.
→ Summarize data with Basic statistical functions.

④ Tables: Convert data into an Excel table or Pivot Table for structured data management.
→ Organize data into sortable, filterable tables.

⑤ Sparklines: Add small in-cell charts like line, column or win/loss to show trends.
→ Display mini charts to visualize trends alongside your data.

⑥ Filters: Apply filters directly from Quick Analysis Tool to narrow down data.
→ Focus on specific part of dataset by filtering values

Lookup functions in EXCEL:

→ These are used to search for a value in the range & return a related value from another range.

(1) VLOOKUP: Vertical Lookup.

→ Searches for a value in the first column of a table & returns a value from another column in the same row.

→ Syn: =Vlookup(lookup_value, table_array, col_index_num, [range_lookup])

→ Ex: =Vlookup(102, A2:C5, 3, False)
to find marks of std with roll no (102)

(2) HLOOKUP: Horizontal Lookup.

→ Searches for a value in the first row of a table & returns a value from another row in the same column.

→ Syn: =Hlookup(lookup_value, table_array, row_index_num, [range_lookup])

→ Ex: =Hlookup(102, B1:D5, 4, False)
to find marks of std with rollno(102)

(3) Index & match:

→ Combining Index & match for more flexible lookup.

→ Index returns the value of a cell within a given range.

→ match returns the relative position of a lookup value.

→ Syn: =Index(array, row_num, [col_num])
=match(lookup_value, lookup_array, [match_type])

→ Ex: =Index(A2:C5, 3, 2)
=match("Ram", B2:B5, 0)

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(4) XLOOKUP:

→ modern lookup.
→ Searches for a range or an array for a specified value & returns the related value from another column.
→ It can lookup both vertically & horizontally & perform exact match, approximate match or wildcard match.
→ Syn: XLOOKUP (look-up-value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode]).

→ Ex: XLOOKUP (103, A2:A5, C2:C5)
find std(103) marks.
→ [match_mode]: 0 → exact match
-1 → exact / next smaller match
+1 → exact / next largest match.
→ [search_mode]:
+1 → search first-to-last
-1 → search last-to-first.

(5) Filter:

→ It is used to extract data that meets specified criteria from a range or table.
→ useful for dynamic filtering based on user inputs
→ Syn: =filter (array, include, [if_empty]).
=(filter (A2:B6, B2:B6 >90, "No result")
marks >90.

Pivot Tables:

→ Powerful excel feature used to summarize, analyze, explore & present large amounts of data.
→ It allows you to quickly recognize & calculate data without changing the original dataset.

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⇒ Steps to create pivot table:

(1) Select the Data Range:
→ ensure data is in table format.
→ Select range of data you want to analyze.

(2) Insert a Pivot Table:
→ Go to insert tab on ribbion
→ Click on PivotTables.
→ Choose whether to place PivotTable in a new worksheet or in existing one.

(3) Configure Pivot table:
→ In pivot table Field list you'll see all data fields
→ Drag those into 4 areas:
→ filters: to filter whole table.
→ columns: what you want in top of table.
→ Rows: what you want listed on the left side
→ Value: numbers you want to calculate.

(4) Customizing PivotTable:
→ you can add / remove field anytime.
→ change arrangement of rows & columns to see data in different ways.

(5) Analyzing Data with Pivot Column:
(i) Summarizing Data: In the Values area you can quickly see totals, counts or averages.
(ii) Grouping data: you can group things together like months into quarters or age into ranges.
(iii) Filtering Data: use filters to focus on what you need like one product or region.
(iv) Formatting: You can apply styles, bold headings or number formats to make PivotTable easy to read.

Ex of PT: Sales summary, Emp analysis, Financial report.

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Difference b/w pivot table & Table:

Table	Pivot Table
(i) used to organize & manage data.	(i) used to summarize, analyze & explore data.
(ii) These are static Data sets.	(ii) These are dynamic data sets can & pull from different data sources.
(iii) Basic formulas & calculations.	(iii) Aggregations (sum, Average, count), etc).
(iv) No dynamic grouping	(iv) Dynamic grouping is possible you can group by categories dates, etc
(v) Auto expands when data is needed.	(v) Requires refresh when data changes
(vi) Simple to setup & use.	(vi) Requires more setup & understanding.
(vii) Limited to sorting & filtering.	(vii) Extensive reporting & analysis capabilities.
(viii) Limited formating & customization.	(viii) Highly customizable, drag & drop interface.

Pivot Table Components:

① **Rows:**
- → Display categories vertically in pivot tables.
- → Use case: grouping data for analysis.
- → Ex: products, Region.

② **Columns:**
- → Display categories horizontally in PT.
- → use case: Comparing accross different groups or time period.
- → Ex: Months, years.

③ **Values:**
- → Contains summarized datas.
- → Ex: sums, average, count.
- → Use case: Summarizing or aggregating numeric data.

④ **Filters:**
- → Allows data to be filtered based on specific conditions.
- → Ex: North, South, East, West for regions.
- → useage: Narrowing down the data displayed in pivot Table.

⑤ **Fieldset:**
- → List of all fields from the dataset to drag into Rows, Columns, Values or Filters
- → usecase: Shaping the PT layout.

⑥ **Slicers:**
- → Visual buttons for quickly filtering datas.
- → usecase: Interactive data filtering.

⑦ **Calculated Field Items:**
- → Custom fields or calculations based on existing data
- → usecase: performing additional calculations.

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Pivot Table Field List:

1. Field List
2. Rows Area
3. Column Area.
4. Values Area.
5. Filter Areas.

Pivot Chart from PivotTable:

→ It is a powerfull tool that provides a visual representation of data summarized in a PivotTable.
→ It allows users to analyze & present large amounts of data dynamically.
→ Changes in PT automatically reflect in pivott chart.
→ Users can easily filter & drill down into data.
→ Provides a clear visual representation of complex data, making it easier to identify trends & patterns.

⇒ Components of Pivot Charts:

1. Chart Area: Encompasses the entire chart, including all elements.
2. Plot Area: Displays the actual data points of series within the chart.
3. Data Series: Represents the actual values plotted on the chart.
4. Legend: Identifies each data series with corresponding colors or patterns.
5. Axes: Provides the framework for chart:
   Xaxis (Horizontal): usually represent categories
   Yaxis (Vertical): usually represent numerical values.
6. Axis Title: Lables that describes what each axis represents.
7. Gridlines: Horizontal/vertical lines that improve readability by acting as reference points.
8. Chart Title: Summarizes the overall focus or themes of the chart.

Pivot Slicers & Timelines:

→ These are interactive tools in Excel used to filter PivotTables dynamically & visually.

⇒ Slicers: allow you to filter pivot Tables data set based on specific categories or fields

→ features:

1. Visual Filtering:
2. Multiselect option.
3. Easy to use
4. customizable

⇒ Timelines: are specialized slicers that allow you to filter PivotTables data based on date field such as years, months, days.

→ features:

1. Time-Based Filtering
2. Interactive Slider.
3. Dynamic Range.
4. Customizable.

Data Visulization:

→ It helps to present data in a graphical format making it easier to understand patterns, trends & insights.

⇒ Types of charts:

1. Bar charts: to compare different categories
2. Line charts: Ideal for showing trend over time
3. Pie charts: Shows proportions of a whole
4. Scatter plot: Displays relationship b/w 2 variables.
5. Histogram: Shows distribution of data.

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Data Validation:

→ It is a feature that helps to control the type of data entered into a cell or range of cells. ensuring data accuracy & consistency.

⇒ features:

① Restrict Input Types:
→ you can limit the type of data that users can input in a cell.

② Create Drop Down lists:
→ you can create drop down lists in cells, so users can select from those options rather than typing.

③ Set Conditions & limits:
→ Can set condition like range of numerical values or date range.

④ Custom Formulas:
→ use formulas to set custom conditions, offering more flexibility for complex data validations.

⑤ Error messages:
→ Define custom error messages that display when user inputs invalid data.

⑥ Input messages:
→ These can be displayed when cell is selected, which provides helpful instructions for users about what data to enter.

Data Consolidation:

→ It involves combining data from multiple sources or worksheets into a single summary.

⇒ Methods:

① Using the Consolidate Feature:
→ Combine data from multiple ranges using various functions (sum, avg, count etc.)

② Manual Consolidation: Manually copy-paste data.

③ Using formulas: use formulas like SUM, AVERAGE & VLOOKUP to combine data.

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→ Data Consolidation Technique:

① Consolidate by position:
→ Consolidate data based on the same position or location across different worksheets.
→ Consolidate data that has same layout.

② Consolidate by category:
→ Consolidate data that has no same layout, but have matching rows/column tables.
→ Consolidate data dynamically, easy to update.

③ Using Pivot Table for consolidation:
→ PT are dynamic way to consolidate data from multiple ranges & summarize it in a interactive way.

④ Using Formulas for Consolidation:
→ Use excel formulas to consolidate data across multiple sheets manually by summarizing each sheet in a summary table.
→ You can use sum, avg, min, max etc.

⑤ Copy & Paste Consolidation:
→ Manual approach for smaller datasets where you can copy data from multiple sheets & paste it into single sheet.
→ Can use functions like sum, subtotal to create summary statistics for combined data.

* Financial Analysis:
→ It involves evaluating the financial health & performance of a business using Excel functions & tools.

① NPV (Net present Value):
→ Calculates present value of a series of ~~the~~ cash flows.
→ Syn:- NPV(rate, value1, [value2], ...)
→ Ex:- NPV(10%, -5000, 1000, 2000)

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(2) IRR [Internal Rate of Return]:
→ Calculates the rate of return for a series of cash flows.
→ Syn: =IRR(values, [guess]).
→ Ex: =IRR(-10000, 3000, 4000, 5000).

(3) PMT [Payment]:
→ Calculates periodic payments for a loan based on constant payments & a constant interest rate.
→ Syn: PMT(rate, nper, pv, [fv], [type]).
nper: Total no. of payment period
pv: present value.
fv: future value.
type: payement at beginning (0)/end(0) of period.
→ Ex: =PMT(5%/12, 60, 10000).

(4) XNPV [Exact Net Present Value]
→ Similar to NPV but accounts for specific dates of cash flows, providing more precise results.
→ Syn: XNPV(rate, values, dates).
→ Ex: =XNPV(-10000, 3000, {"2025-01-01", "2026-01-01"})

(5) XIRR [Exact Internal Rate of Return].
→ Calculates the interest rate of return for a series of periodic cashflow.
→ Syn: =MIRR(values, finance_rate, reinvest_rate)
→ Ex: =MIRR({-10000, 3000, 4000, 5000}, 5%, 8%).

(6) DB: [Declining Balance Depreciation].
→ Calculates depreciation of an asset for a specified period using fixed declining balance method.
→ Syn: DB(cost, salvage, life, period, [month])
→ Ex: =DB(1000, 1000, 5, 1).

(8) SLN: (Straight line Depreciation).
→ Calculates straight line depreciation of an asset for one period.
→ Syn: =SLN(cost, salvage, life)
→ Ex: =SLN(10000, 1000, 5).

(9) FV: [Future Value]:
→ Calculates future value of an investment based on periodic, constant payments & a constant interest rate.
→ Syn: =FV(rate, nper, pmt, [pv], [type])
→ Ex: =FV(5%/12, 60, -200).

(10) PV (Present Value):
→ Calculates present value of an investment/loan based of series of future payments & a constant interest rate.
→ Syn: =PV(rate, nper, pmt, [fv], [type])
→ Ex: =PV(5%/12, 60, -200).

What-if analysis:
→ It is a tool that allows you to try different values for formulas in worksheet to determine the impact of those changes on the results.
→ Types:
(1) Scenario managers:
→ Create & manage different scenarios.
→ Compare the impact of different inputs on data
(2) Goal seek:
→ Find necessary input value to achieve desired result.
(3) Data Tables:
→ Explore the impact of different variables on the data.
→ Changing 1 or 2 variables effect outputs.

Importing data into EXCEL:

→ You can import data from different sources such as text, CSV files, database & web pages.

→ Steps:

1. Select Data Source.
2. Import the data
3. Clean & organize data.

Data Model:

→ It allows combining data from multiple tables, creating relationships & building more complex analyses.

→ Creating a Data Model:

1. Add data to the Data model.
2. Create Releadionship.
3. Build PivotTable.

→ Ex: Sales & customers, Inventory & orderss

Power Pivot:

→ It is an advance data Analysis tool that allows working with large databases & creating complex data models.

→ Using Power pivot:

1. Adding Data to power pivot.
2. Creating Relationship.
3. Building PivotTables.

→ Ex: Sales & Marketing, Financial Reporting.

Power View: Ex: Sales dashboard, customer analysis.

→ It is a data visualization Tool that allows creating interactive reports & dashboards.

→ Creating power view:

1. Add data to power view.
2. Create visualizations.
3. Interactive features.

* Aesthetic Power View Reports:

⟹ It makes power view report more visually appealing & easier to interpret.

→ Design:

① use Themes & styles.
② Adding Images & Logos.
③ Customizing Layouts.

* Key Performance Indicators KPI's:

→ these are quantifiable measures used to evaluate how effectively an individual group or organization is achieving key objective.

→ Characteristics of KPI's:

① Specific: Clearly defined & focused.
② measurable: Expressed numerically.
③ Relavent: Directly linked to goals.
④ Time based: can be tracked over time.

⇒ Designing:

① Prepare the Data
② Define KPI Targets. Ex: Avg Grade ≥ 70%,
attendance ≥ 85%,
Pass % ≥ 90%.
③ Calculate KPI's using Excel formulas.
④ Apply Conditional formatting.

⇒ Ex: Std evalutions, Business analysis, Emp analysis